KCC CODE CLUB · GAME MAKERS

Week 3

Loops and Randomness

Week 3 of 8

Name _____

Today's big idea

If you're about to do the same thing twenty times — **don't**. Tell the computer to repeat it.

```
repeat 4                                   CONTROL
    clap
    clap
    stomp
```

That's 12 actions written in 4 lines. Change the 4 to a 100 and you've just done 300 actions by typing two characters.

And when you want the computer to surprise you, use `pick random`.

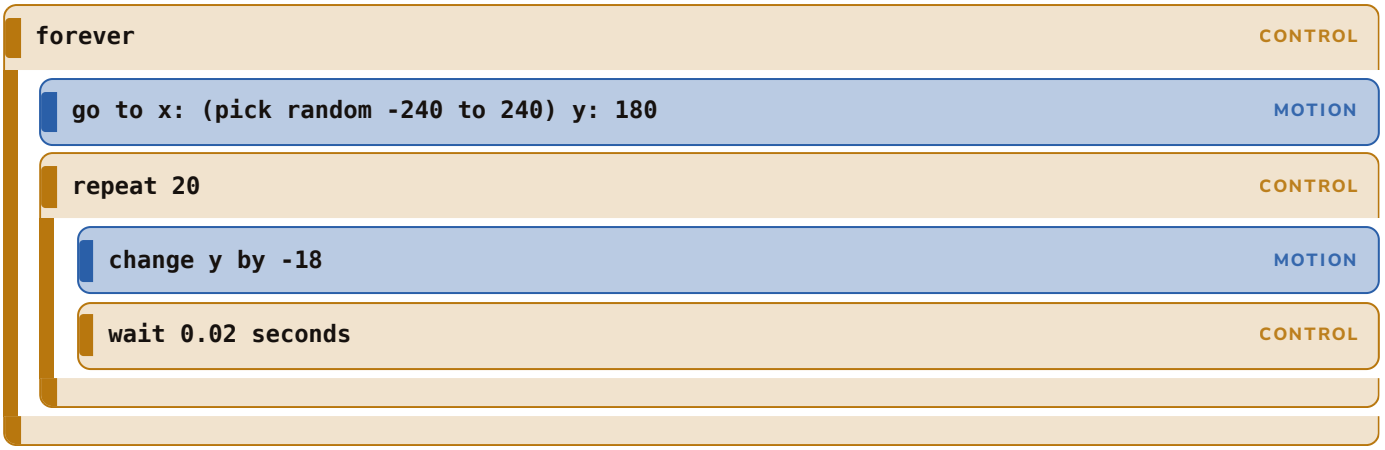
Words to know

Word	What it means
loop	Code that repeats
repeat n	Do this exactly n times
forever	Do this until the game stops. Nothing can go after it
random	A number the computer picks that you can't predict

> **Golden rule:** every `forever` loop needs a `wait` inside it somewhere. Without one, Scratch freezes and you lose your work.

Class activity — Rain

```
when green flag clicked                    EVENTS
```



A raindrop that appears somewhere random along the top and falls. Forever.

Now change the numbers. Fewer repeats + bigger step = faster. Try it.

Your project — things fall from the sky!

Today your project starts looking like a real game.

- [ ] Add a falling object sprite that fits your theme
- [ ] Make it the right size
- [ ] Give it the falling script (same shape as Rain)
- [ ] **Tune the speed** so it's catchable but not boring
- [ ] **Save it**

It doesn't notice your player yet — you can "catch" it and nothing happens. That's next week.

Homework

1. A second thing to catch. Add another falling sprite that looks different. *(Shortcut: right-click your first one and duplicate it — the code comes with it!)*

2. Unpredictable speed. Use `pick random` so each drop falls at a different speed. *(Hint: the `wait` can take a random number.)*

Bonus challenges

1. Draw shapes with `repeat` + `turn`. A square is `repeat 4 [move 100, turn 90]`. What makes a triangle? A hexagon? What does `repeat 36 [move 10, turn 10]` draw?
2. Make your falling object spin as it falls.
3. **Dancing Sprite.** Pick a sprite with a walk cycle. `forever` → `next costume`, `wait 0.2 seconds`, `move 5 steps`. Tune the wait until the walk looks right, then add `if on edge, bounce`.
4. Use `repeat until` instead of `forever` so it stops after ten drops. *(Hard! You'll need a way to count, which we learn in two weeks.)*

If something breaks

- **Flickers at the top and nothing else?** No `wait` inside the falling loop.
- **Falls once then stops?** The whole thing needs to be inside a `forever`.
- **Falls upward?** `change y by 18` should be `change y by -18`. Up is positive.
- **Scratch froze?** A `forever` with no `wait` in it. Reload and add one.

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.